

Stanislau Beliasau

Green Card holder

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Research Engineer combining strong research intuition with production-focused systems engineering. Track record includes developing novel approaches that anticipate published research directions, scaling frontier generative models on large GPU clusters, and optimizing inference. Expertise spans distributed training, model architecture design, and deployment optimization across the full ML stack.

Work Experience

  **Mirage (FKA Captions)**, New York, USA

Research Engineer, Member of Technical Staff

July. 2023 – Present.

- **Pre-trained Mirage, a 13B parameter transformer-based text+audio+image-to-video generation model, from scratch on massive multimodal datasets.** Designed and implemented distributed training infrastructure on 1000+ H100 GPU cluster, achieving 40% MFU, implemented context-parallel training enabling sequence lengths up to 150k tokens.
- Optimized training stability and convergence: tuned optimizer hyperparameters (learning rate schedules, warmup strategies, weight decay, batch sizes), implemented gradient clipping and loss scaling strategies, and developed checkpoint recovery mechanisms for fault-tolerant training at scale.
- Designed novel architectural components for multimodal transformer through extensive ablation studies and architecture exploration. Conducted scaling experiments to optimize model performance and training efficiency trade-offs.
- Built automated evaluation and monitoring infrastructure for pre-training: designed pub/sub system for continuous checkpoint assessment, integrated distributed metrics computation, Weights&Biases logging, and automated artifact uploads to cloud storage for tracking training progress.
- Optimized Mirage Audio pre-training (10B parameter text+audio-to-audio transformer), achieving 30% training throughput improvement.
- Optimized Mirage Video inference achieving 4x faster startup times and 2x overall speedup through fp8 quantization, custom attention kernels, torch compilation, and inference-time caching with minimal quality degradation.
- Optimized Mirage Audio inference achieving 5x faster startup times and 4x inference speedup through efficient kernels, torch compilation while maintaining quality.
- Designed and implemented audio-to-landmarks architecture for lip-sync generation through systematic architecture exploration and ablation studies.
- Evaluated GPU providers and made key infrastructure decisions enabling efficient training at scale.

  **Gradient & Persona: AI Photo & Video mobile editors**, New York, Remote

Computer Vision Engineer

Aug. 2022 – May. 2023

- Designed and implemented novel image encoding method for personalized Stable Diffusion generation, replacing expensive DreamBooth fine-tuning with single-shot encoding from few images. Approach enabled identity-preserving generation without model fine-tuning; similar methods were later published and widely adopted in 2024.
- Optimized Stable Diffusion inference achieving 30% speedup through architectural modifications, custom modules, and efficient sampling strategies.
- Trained production models for re-aging, body reshaping, and segmentation through rapid architecture exploration and dataset curation strategies.

 **Yandex School of Data analysis**, Remote

ML course tutor assistant

Feb. 2022 – June. 2023

- Served as ML course tutor at Yandex School of Data Analysis, giving lectures on NLP and computer vision, and designing coursework on distributed LLM training (GPT-2XL, OPT-6.7B).

 **iTechArt**, Minsk, Belarus

Machine Learning Engineer

Nov. 2021 – Aug. 2022

 **Yandex**, Minsk, Belarus

Software Engineer Intern

May. 2021 – Nov. 2021

Publications

Seeing Voices: Generating A-Roll Video from Audio with Mirage: *arXiv:2506.08279*, 2025. [\[link\]](#)

Shaping voices: Crafting expressive audio: *blog post*, 2025. [\[link\]](#)

How to Fine-Tune Very Large Model if It Doesn't Fit on Your GPU: Technical article on distributed training techniques for large language models. 600+ reactions, 2022. [\[link\]](#)

Education

 **Yandex School of Data analysis**, Minsk, Belarus

Master's degree level Machine Learning developer academic program

Sep. 2020 – June. 2022

- Relevant coursework: Efficient Deep Learning Systems, Reinforcement Learning, Computer Vision, NLP, Recommendation Systems.

 **Belarusian State University**, Minsk, Belarus

Bachelor of Computer Science

Sep. 2018 – Aug. 2022

- Awarded a full scholarship and stipend by the government per entrance exam results.